

## **Introduction**

Rental prices near UCLA vary substantially across properties and neighborhoods. This study investigates how monthly rental prices in the UCLA area are associated with property size, number of bedrooms and bathrooms, property type, and geographic location. Understanding these relationships may help renters evaluate whether a listed property is reasonably priced and identify which property characteristics are most strongly associated with rent.

The data were obtained from rental listings collected through RentCast. The cleaned dataset contains 2,212 active long-term rental listings across 15 ZIP codes surrounding UCLA. The variables used in the analysis are monthly rental price, ZIP code, number of bedrooms, number of bathrooms, square footage, and property type. The four retained property types are Apartment, Condo, Single Family, and Townhouse.

A multiple linear regression model will be used to examine the relationship between rental price and the explanatory variables. This method is appropriate because the response variable, monthly rental price, is quantitative, while the dataset contains several potential predictors that may jointly influence price. Multiple regression allows the effect of each property characteristic to be estimated while controlling for the other variables in the model. Bedrooms, bathrooms, and square footage will be treated as quantitative predictors, while ZIP code and property type will be treated as categorical predictors. Model assumptions, including linearity, constant variance, normality of residuals, and independence, will be evaluated using residual and diagnostic plots. Potential outliers, leverage points, and influential observations will also be examined.

The report is organized as follows. The Data Description section presents summary statistics and graphical analyses to examine the distributions of the variables and the relationships among them. The Results and Interpretation section reports the fitted regression models, compares several candidate models, selects the best predictive model among them, and evaluates its assumptions using diagnostic tools. Finally, the Discussion section summarizes the main findings, considers whether the final model is reasonable, and discusses the limitations of the analysis and possible directions for future improvement.

## **Data Description**

### **Sample and Variables**

Our sample contains 2,212 active long-term rental listings collected from the RentCast API, covering 15 zip codes surrounding UCLA: 90019, 90024, 90025, 90034, 90035, 90036, 90048, 90049, 90064, 90066, 90069, 90230, 90403, 90404, and 90405 (see Appendix Table A1 for zip-level counts). The dataset includes six variables:

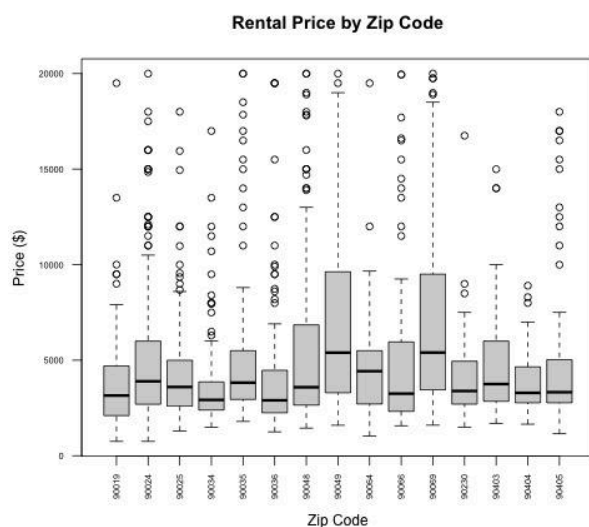
- price — monthly rent in US dollars (response variable)
- zipCode — zip code (categorical, 15 levels)
- bedrooms — number of bedrooms
- bathrooms — number of bathrooms

- squareFootage — unit size in square feet
- propertyType — housing type (Apartment, Condo, Single Family, Townhouse)

The raw merged data initially contained 2,519 listings across 16 fields. We removed non-informative or redundant fields (listing ID, address, city, state, duplicate zip field), dropped yearBuilt due to a high missing rate (31.6%), and tested latitude/longitude as an alternative to zip code — a VIF and partial F-test showed they were highly collinear with zip code (VIF = 11.4 and 18.0) and added only a marginal, though statistically detectable, improvement in fit, so we kept zipCode alone for interpretability. We also removed listings with missing squareFootage/bathrooms, a property type category with too few observations to estimate reliably (Multi-Family,  $n = 3$ ), and listings with implausible price-per-square-foot ratios (below \$1 or above \$20 per sq ft, consistent with the standard used in Mini Assignment 2), which were consistent with data entry errors. This left the sample of 2,212 observations. (Full step-by-step justification is provided in the Discussion section.)

### Summary Statistics

| Variable             | Min | Q1    | Median  | Mean     | Q3       | Max    | SD       |
|----------------------|-----|-------|---------|----------|----------|--------|----------|
| price (\$)           | 750 | 2,650 | 3,600   | 4,881.22 | 5,661.25 | 20,000 | 3,606.32 |
| bedrooms             | 0   | 1     | 2       | 1.88     | 3        | 14     | 1.22     |
| bathrooms            | 1   | 1     | 1.5     | 1.81     | 2        | 8      | 0.99     |
| squareFootage (sqft) | 90  | 700   | 1,015.5 | 1,192.85 | 1,450.25 | 5,505  | 723.84   |



Price is strongly right-skewed (mean \$4,881 vs. median \$3,600), with a long tail of high-end listings up to \$20,000/month; squareFootage shows a milder version of the same pattern. As shown below, price varies substantially by both location and property type: mean price ranges from \$3,472 in 90034 (Palms) (Appendix Table A1 for Mean price by zip code) to \$7,087 in 90049 (Brentwood), and Single Family homes (\$7,727 average) rent for more than double the Apartment average (\$3,230), with Condo and Townhouse in between — both patterns match real-world expectations. All three continuous predictors are positively correlated with price ( $r = 0.64$ – $0.85$ ) and with each other ( $r = 0.76$ – $0.80$ ), foreshadowing a multicollinearity check in the modeling section (full correlation matrix and scatterplot matrix in Appendix *Table & Figures A*).

## Results and Interpretation

### Initial Multivariable Regression Model

After applying the stated cleaning criteria, one additional listing was removed because it reported 14 bedrooms and 8 bathrooms within 125 square feet, a physically implausible combination that could not be verified. The resulting analytic sample contained 2211 listings.

We first fitted an untransformed multiple linear regression model using monthly rental price as the response variable and ZIP code, bedrooms, bathrooms, square footage, and property type as predictors. ZIP code 90019 and Apartment were used as the reference categories. The initial model explained approximately 78.25% of the variation in monthly rental price, with an adjusted  $R^2$  of 0.7805. The overall regression was statistically significant,  $F(20,2190)=393.8$ ,  $p<0.001$ , indicating that the predictors collectively provided substantial explanatory power for rental price.

Holding the other variables constant, bathrooms and square footage were significant positive predictors of monthly rent. One additional bathroom was associated with an estimated increase of approximately \$452 per month, while one additional square foot was associated with an estimated increase of approximately \$3.59 per month. Bedrooms were not statistically significant at the 5% level in the initial model. This suggests that, after controlling for total square footage, bathrooms, location, and property type, the number of bedrooms did not provide substantial additional explanatory information in the untransformed model. (See summary of the initial full model in Appendix *Table 2.1*)

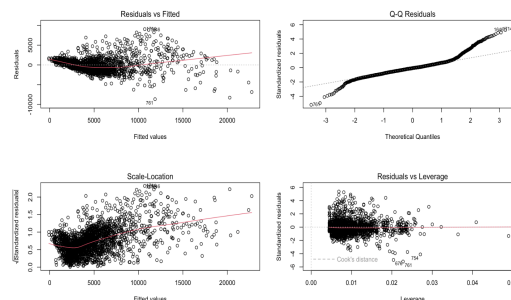


Figure 2.1

Although the initial model had a relatively high  $R^2$ , the diagnostic plots indicated several violations of the linear regression assumptions. The Residuals vs. Fitted plot showed systematic curvature, suggesting that the relationships between rent and one or more predictors were not adequately represented by a purely linear specification. The Scale-Location plot showed increasing residual variability as fitted rent increased, providing evidence of heteroscedasticity. In addition, the Normal Q-Q plot showed noticeable departures from normality in both tails, while the Residuals vs. Leverage plot identified several potentially unusual or influential observations. These results indicated that transformations were to be considered before conducting variable selection.

Multicollinearity was also assessed using generalized variance inflation factors. Although bedrooms, bathrooms, and square footage were strongly correlated in the exploratory analysis, all adjusted GVIF values were approximately 2 or lower. Therefore, there was no evidence of severe multicollinearity that would require removing predictors from the model. (See the GVIF results in Appendix *Table 2.2*).

### **Transformation of model variables**

To identify an appropriate transformation of monthly rental price, we examined both the inverse response plot and the Box-Cox profile likelihood plot. The inverse response plot suggested a transformation parameter of approximately  $\lambda=0.6$ , which is reasonably close to a square-root transformation, whereas the Box-Cox analysis suggested a value near  $\lambda=0$ , corresponding to a logarithmic transformation. Based on these results, two candidate response transformations were considered. (See the two plots in appendix *Figure 2.2 & 2.3*)

Neither response transformation completely eliminated the departures from normality and constant variance. However, the model using  $\log(\text{price})$  produced a slightly more stable residual spread and somewhat less curvature in the residual plots than the model using  $\sqrt{\text{price}}$ . The logarithmic transformation was also directly supported by the Box-Cox result. Therefore,  $\log$ -transformed monthly rent was selected as the response variable for the remaining candidate models. (See the regression output and diagnostics in Appendix *Tables 2.3 & 2.4*, *Figures 2.4 & 2.5*.)

Although the  $\log$  transformation substantially improved the behavior of the response variable, the diagnostic plots still showed some systematic residual patterns. To investigate whether these patterns were associated with the continuous predictors, standardized residuals were plotted separately against bedrooms, bathrooms, and square footage.

Some curvature was visible for all three predictors. However, square footage is continuous and has observations distributed across a much wider range. Its residual plot showed a clearer downward pattern at larger values, suggesting that the relationship between  $\log(\text{price})$  and raw square footage remained nonlinear. (See residual plots in appendix *Figure 2.6*). For this reason, square footage was considered first for further transformation. Two candidate models were fitted:

*Model2: log(price) ~ ZIP + bedrooms + bathrooms + log(squareFootage) + propertyType*  
*Model3: log(price) ~ ZIP + bedrooms + bathrooms + sqrt(squareFootage) + propertyType*

The diagnostic plots for the logarithmic and square-root transformations of square footage were generally similar, and neither model completely removed all departures from the regression assumptions. Therefore, AIC and BIC were used to compare the two models.

The model using  $\sqrt{\text{squareFootage}}$  had an AIC of 335.99 and a BIC of 461.42, whereas the model using  $\log(\text{squareFootage})$  had an AIC of 425.86 and a BIC of 551.28. The substantially lower AIC and BIC values favored the square-root square-footage model. Its adjusted  $R^2$  was also slightly higher, approximately 0.805 compared with approximately 0.797 for the logarithmic square-footage model. (See results in appendix *Table 2.5 & 2.6*)

Therefore, among the candidate models, the selected final model was:

*log(price) ~ ZIP + bedrooms + bathrooms + sqrt(squareFootage) + propertyType*

### **Variable selection**

Forward and backward stepwise selection based on AIC both returned the same set of predictors as the transformed full model. Therefore, the variable-selection procedures provided no evidence that ZIP code, bedrooms, bathrooms, square footage, or property type should be removed.

The final model explained approximately 80.7% of the variation in log monthly rent, with an adjusted  $R^2$  of approximately 0.805. The overall regression was statistically significant,  $F(20,2190)=456.9$ ,  $p<0.001$ . Bedrooms, bathrooms, square-root square footage, ZIP code, and property type were retained in the final model. (See summary of final model in appendix *figure 2.7*)

### **Outliers, leverage points, and influential observations**

Using conventional screening thresholds, 28 observations had absolute standardized residuals greater than 3, 103 exceeded the leverage threshold of  $2p/n$ , and 110 exceeded the Cook's distance threshold of  $4/n$ . However, the maximum Cook's distance was only 0.022, indicating that no single observation exerted exceptionally strong influence on the fitted model. Therefore, these observations were retained

### **Discussion**

Square-root-transformed square footage was the strongest continuous predictor in the final model. Because the response variable was log-transformed, the coefficients can be interpreted approximately as percentage changes in monthly rent, holding the other predictors constant. A one-unit increase in  $\sqrt{\text{squareFootage}}$  was associated with an estimated 4.1% increase in monthly rent. Because square footage enters the model through a square-root

transformation, the model implies diminishing marginal effects of additional raw square footage. For example, increasing unit size from 900 to 1,600 square feet increases  $\sqrt{\text{squareFootage}}$  from 30 to 40 and corresponds to an estimated increase of approximately 50% in predicted monthly rent. Bedrooms and bathrooms were also significant positive predictors. Holding square footage and the other variables constant, each additional bedroom was associated with approximately 3.9% higher predicted rent, while each additional bathroom was associated with approximately 5.8% higher predicted rent. This may reflect differences in layout, functionality, or desirability that are not captured by total floor area alone.

Property type and zip code continue to capture a substantial portion of the remaining variation. Relative to apartments, single-family homes have an 18.4% premium, while condos have a 19.5% premium, and townhouses have a 13.3% premium. This indicates to us that even after controlling for the physical characteristics of a unit, the type of housing still carries an independent value. This is likely reflecting privacy, amenities, or perceived exclusivity not captured by our other variables. The effect of location is even larger. 90069 (West Hollywood) shows the largest premium of the entire sample, at roughly 62% higher rent than that of the reference zip code (90019). Furthermore, zip codes such as 90035 and 90230 show more modest premiums in the range of 11-16%. This suggests that the neighborhood carries substantial explanatory power independent of what the unit's size or type is.

Our findings are reasonable and consistent with what we would expect regarding the West LA rental market. Square footage being one of the most influential factors aligns with the basic idea that the amount of space is a highly important factor for determining the rent. The influence of zip codes also aligns with the idea that location and perceptions of said location also have a role in determining rent, as people would want to live somewhere more if they have a better perception of it regardless of unit size. These results can enable renters to have a reasonable benchmark of whether a listed price is fair, given the size, room count, type, and its ZIP code. It could also help property managers with creating fair pricing or knowing what to prioritize when renovating in order to maximize its fair rental price.

Some limitations of our model lie in the diagnostic checks that we did. Despite choosing to transform using the square root transformation, there is still some heteroscedasticity, non-normality in the tails, and some slight non-linearity remaining in the residual plots. Our model also fails to capture more qualitative property features, such as how old the building is, amenities nearby (such as access to public transportation, parking, in-unit laundry), or the renovation quality. That most likely explains some of the still unexplained variance. Some future work that we could do is accounting for these additional variables, exploring the interaction between them, and working on finding some more essential predictors for estimating the rent.

## Appendix

*Table&Figures A1:*

### Categorical variable distributions:

| zipCode | n   | zipCode | n   | propertyType  | n     |
|---------|-----|---------|-----|---------------|-------|
| 90019   | 182 | 90048   | 176 | Apartment     | 1,076 |
| 90024   | 206 | 90049   | 128 | Condo         | 388   |
| 90025   | 254 | 90064   | 100 | Single Family | 538   |
| 90034   | 212 | 90066   | 122 | Townhouse     | 210   |
| 90035   | 186 | 90069   | 249 |               |       |
| 90036   | 130 | 90230   | 48  |               |       |
|         |     | 90403   | 89  |               |       |
|         |     | 90404   | 58  |               |       |
|         |     | 90405   | 72  |               |       |

### Mean price by zip code:

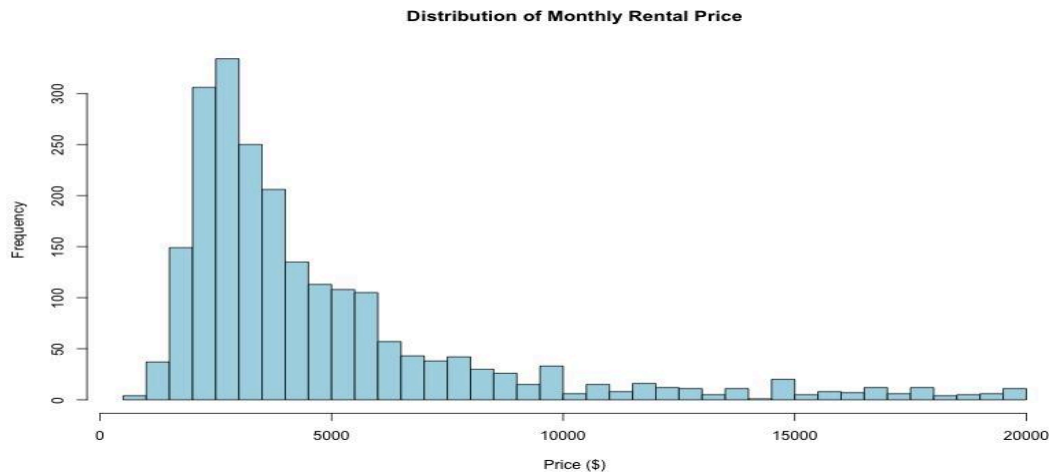
| Zip                    | Mean price (\$) | Zip                  | Mean price (\$) |
|------------------------|-----------------|----------------------|-----------------|
| 90049 (Brentwood)      | 7,087.34        | 90035                | 4,823.58        |
| 90069 (West Hollywood) | 6,889.50        | 90403 (Santa Monica) | 4,727.47        |
| 90048                  | 5,435.31        | 90064                | 4,698.19        |
| 90405 (Santa Monica)   | 5,127.61        | 90025 (West LA)      | 4,145.71        |
| 90024 (Westwood)       | 5,067.97        | 90230                | 4,183.56        |
| 90066                  | 4,862.04        | 90036                | 4,097.15        |
|                        |                 | 90404                | 3,849.10        |
|                        |                 | 90019                | 3,726.35        |
|                        |                 | 90034 (Palms)        | 3,471.62        |

### Correlation Matrix (continuous variables):

|       | price | bedrooms | bathrooms | squareFootage |
|-------|-------|----------|-----------|---------------|
| price | 1.000 | 0.641    | 0.704     | 0.853         |

|               |       |       |       |       |
|---------------|-------|-------|-------|-------|
| bedrooms      | 0.641 | 1.000 | 0.804 | 0.764 |
| bathrooms     | 0.704 | 0.804 | 1.000 | 0.797 |
| squareFootage | 0.853 | 0.764 | 0.797 | 1.000 |

## Distributions and Relationships



**Price** is strongly right-skewed: the mean (\$4,881) is noticeably higher than the median (\$3,600), reflecting a long right tail driven by a relatively small number of high-end listings (up to \$20,000/month).

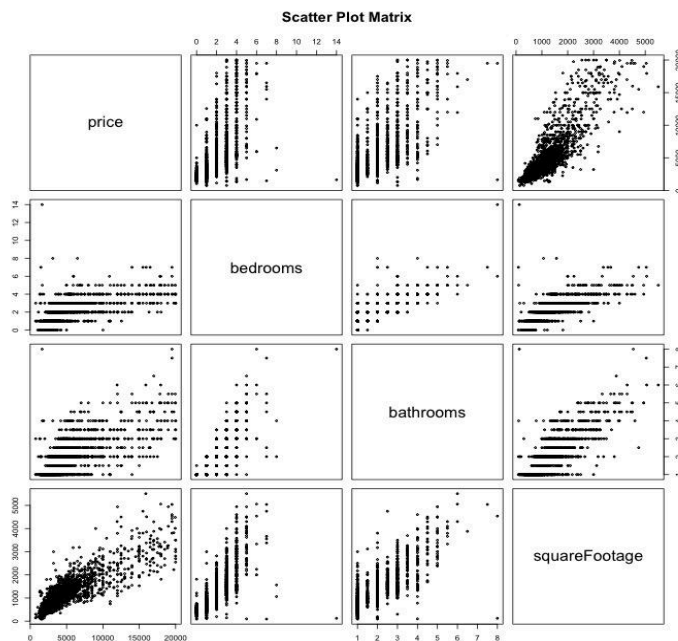
**squareFootage** shows a milder version of the same pattern (median 1,015.5 sq ft, mean 1,192.85 sq ft). **bedrooms** and **bathrooms** are more moderately distributed (median 2 and 1.5, respectively), with a small number of unusually large properties (up to 14 bedrooms, 8 bathrooms) pulling the upper tail.

**Price by zip code** varies substantially across the study area, from \$3,472 in 90034 (Palms) to \$7,087 in 90049 (Brentwood) — a pattern broadly consistent with the known relative affordability of these Los Angeles neighborhoods, which lends some external validity to the raw data before any modeling adjustments.

**propertyType distribution:** Apartment is the most common type (1,076 of 2,212 listings, about 49%), followed by Single Family (538, ~24%), Condo (388, ~18%), and Townhouse (210, ~9%).

Price by property type also varies substantially: Apartment listings are the cheapest on average (\$3,230), while Single Family homes command the highest average price (\$7,727) — more than double the Apartment average — with Condo (\$5,411) and Townhouse (\$5,073) in between. This ordering matches real-world expectations, since single-family homes tend to be larger and more private than apartments.

**Relationships among the continuous predictors:** all three are positively correlated with price, with squareFootage showing the strongest association ( $r = 0.853$ ), followed by bathrooms ( $r = 0.704$ ) and bedrooms ( $r = 0.641$ ) — consistent with the intuitive expectation that larger units command higher rents. However, squareFootage, bedrooms, and bathrooms are also highly correlated with one another ( $r$  ranging from 0.764 to 0.804). This pairwise correlation among predictors is worth flagging early, since it foreshadows a multicollinearity (Variance Inflation Factor) check in the modeling section.



The four generated figures confirm the numbers above: the histogram shows a sharp peak around \$2,500–\$3,000 with a long right tail out to \$20,000; the zip code boxplot shows that nearly every zip code has a cluster of high-price outliers above its whiskers, with 90049 and 90069 visibly having the highest medians and widest boxes, consistent with their leading mean prices; the property type boxplot shows Apartment has the lowest and tightest box (with many high-price outliers above it), while Single Family has the widest box and highest median, consistent with its highest mean price; and the scatterplot matrix shows the price–squareFootage relationship is the tightest and most linear-looking of the three continuous predictors, while bedrooms and bathrooms (being discrete, low-cardinality variables) show more banding/overlap, consistent with their lower correlations with price.

## Results and interpretation

```
Call:
lm(formula = price ~ zipcode + bedrooms + bathrooms + squareFootage +
    property_type, data = my_data1_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-8662.2  -912.7   -99.7    709.5   9050.5

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.502e+03  1.442e+02 -10.415 < 2e-16 ***
zipcode90024  1.353e+03  1.747e+02  7.748 1.42e-14 ***
zipcode90025  6.243e+02  1.649e+02  3.787 0.000157 ***
zipcode90034  9.405e+02  1.718e+02  5.475 4.88e-08 ***
zipcode90035  3.464e+02  1.768e+02  1.960 0.050146 .
zipcode90036  8.894e+02  1.967e+02  4.521 6.47e-06 ***
zipcode90048  1.456e+03  1.797e+02  8.103 8.79e-16 ***
zipcode90049  1.308e+03  1.988e+02  6.577 5.98e-11 ***
zipcode90064  5.222e+02  2.138e+02  2.443 0.014660 *
zipcode90066  1.178e+03  1.985e+02  5.935 3.40e-09 ***
zipcode90069  2.651e+03  1.673e+02  15.848 < 2e-16 ***
zipcode90230  6.578e+02  2.748e+02  2.394 0.016768 *
zipcode90403  1.652e+03  2.202e+02  7.500 9.21e-14 ***
zipcode90404  1.121e+03  2.556e+02  4.385 1.21e-05 ***
zipcode90405  2.253e+03  2.359e+02  9.548 < 2e-16 ***
bedrooms      -8.567e+01  5.724e+01 -1.497 0.134608
bathrooms     4.557e+02  7.154e+01  6.371 2.29e-10 ***
squareFootage  3.574e+00  9.977e-02  35.821 < 2e-16 ***
property_typeCondo  4.926e+02  1.040e+02  4.697 2.80e-06 ***
property_typeSingle Family  9.654e+02  1.043e+02  9.253 < 2e-16 ***
property_typeTownhouse  5.645e+01  1.316e+02  0.429 0.667995
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1690 on 2190 degrees of freedom
Multiple R-squared:  0.7824,    Adjusted R-squared:  0.7804
F-statistic: 393.8 on 20 and 2190 DF,  p-value: < 2.2e-16
```

Table 2.1

|               | GVIF     | Df | GVIF^(1/(2*Df)) |
|---------------|----------|----|-----------------|
| zipcode       | 1.315736 | 14 | 1.009848        |
| bedrooms      | 3.605927 | 1  | 1.898928        |
| bathrooms     | 3.833188 | 1  | 1.957853        |
| squareFootage | 4.033672 | 1  | 2.008400        |
| property_type | 1.490916 | 3  | 1.068831        |

Table 2.2

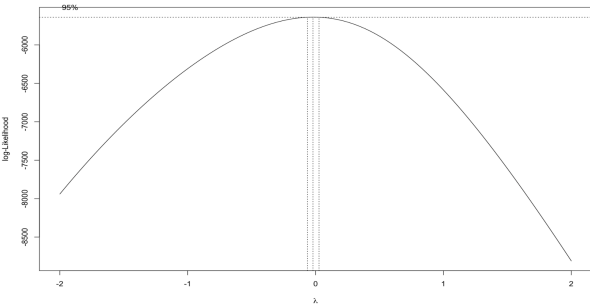


Figure 2.2

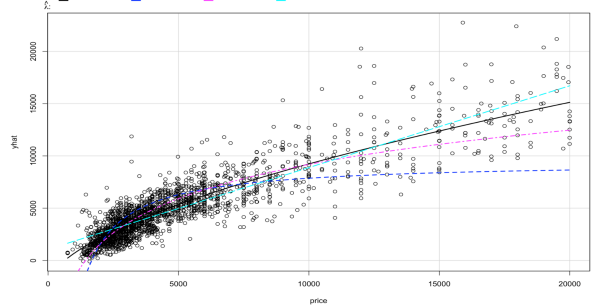


Figure 2.3

```
Call:
lm(formula = sqrt(price) ~ zipcode + bedrooms + bathrooms + squareFootage +
    property_type, data = my_data1_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-47.898  -5.409   -0.887   4.512  43.855

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.585e+01  0.285e+01  31.178 < 2e-16 ***
zipcode90024  9.365e+00  1.001e+00  9.334 < 2e-16 ***
zipcode90025  4.962e+00  9.470e-01  5.240 1.76e-07 ***
zipcode90034  6.156e+00  9.866e-01  6.130 6.30e-08 ***
zipcode90035  3.253e+00  1.811e+00  1.804 0.081377 **
zipcode90036  5.773e+00  1.130e+00  5.063 4.44e-07 ***
zipcode90048  9.454e+00  1.832e+00  5.157 < 2e-16 ***
zipcode90049  9.174e+00  1.142e+00  8.034 1.53e-15 ***
zipcode90064  5.130e+00  1.223e+00  4.161 3.73e-05 ***
zipcode90066  7.512e+00  1.140e+00  6.588 5.58e-11 ***
zipcode90069  1.776e+01  9.680e-01  18.461 < 2e-16 ***
zipcode90230  4.686e+00  1.579e+00  2.964 0.003036 **
zipcode90403  1.155e+01  1.705e+00  6.770 < 2e-16 ***
zipcode90404  8.872e+00  1.468e+00  6.049 4.30e-08 ***
zipcode90405  1.273e+01  1.355e+00  9.423 < 2e-16 ***
bedrooms     1.235e+00  3.288e-01  3.755 0.000178 ***
bathrooms    2.334e+00  1.180e-01  19.825 6.54e-09 ***
squareFootage 1.955e+02  5.731e-04  34.115 < 2e-16 ***
property_typeCondo  5.585e+00  6.024e-01  9.272 < 2e-16 ***
property_typeSingle Family  6.130e+00  9.391e-01  65.529 < 2e-16 ***
property_typeTownhouse  3.122e+00  7.559e-01  4.130 3.77e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.787 on 2190 degrees of freedom
Multiple R-squared:  0.8021,    Adjusted R-squared:  0.8011
F-statistic: 446.2 on 20 and 2190 DF,  p-value: < 2.2e-16
```

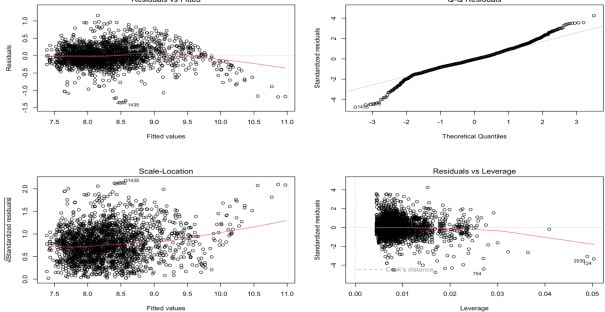


Table 2.3

Figure 2.4

```
Call:
lm(formula = log(price) ~ zipcode + bedrooms + bathrooms + squareFootage +
    property_type, data = my_data1_clean)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.29136 -0.13744 -0.00894  0.15504  1.13182
```

```
Coefficients:
(Intercept)      7.177e+00  2.338e-02  307.219 < 2e-16 ***
zipcode00024    -2.785e-01  2.829e-02  9.846 < 2e-16 ***
zipcode00025    -1.709e-01  2.678e-02  6.402 3.87e-10 ***
zipcode00034    -1.831e-01  2.752e-02  6.582 5.81e-11 ***
zipcode00035    -1.305e-01  2.863e-02  4.559 5.42e-06 ***
zipcode00036    -1.031e-01  3.166e-02  5.120 3.33e-07 ***
zipcode00048    -2.733e-01  2.911e-02  9.388 < 2e-16 ***
zipcode00049    -2.038e-01  3.220e-02  8.814 < 2e-16 ***
zipcode00064    -1.933e-01  3.462e-02  5.583 2.66e-08 ***
zipcode00066    -2.137e-01  3.215e-02  6.648 3.75e-11 ***
zipcode00069    -5.351e-01  2.709e-02 19.400 < 2e-16 ***
zipcode00230    -1.547e-01  4.451e-02  3.475 0.00052 ***
zipcode00403    -3.544e-01  3.557e-02  9.936 < 2e-16 ***
zipcode00404    -2.001e-01  4.148e-02  6.284 3.97e-10 ***
zipcode00405    -3.720e-01  3.811e-02  9.709 < 2e-16 ***
bedrooms         8.333e-02  9.278e-03  8.987 < 2e-16 ***
bathrooms        4.797e-02  1.159e-02  4.140 3.40e-05 ***
squareFootage    -4.391e-04  1.616e-05 28.411 < 2e-16 ***
property_typeCondo 2.170e-01  1.008e-02 12.774 < 2e-16 ***
property_typeSingle 1.120e-01  1.608e-02 10.179 < 2e-16 ***
property_typeTownhouse 1.573e-01  2.131e-02  7.381 2.22e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.2737 on 2190 degrees of freedom
Multiple R-squared:  0.7853, Adjusted R-squared:  0.7833
F-statistic: 400.2 on 20 and 2190 DF, p-value: < 2.2e-16
```

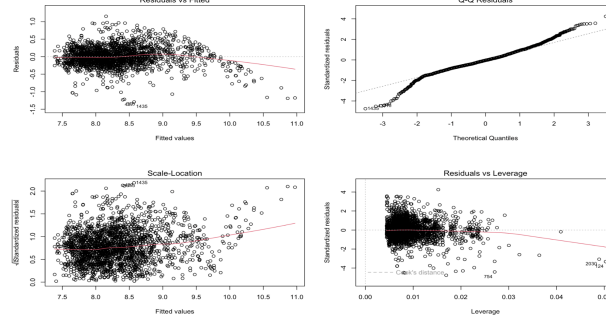


Figure 2.5

Figure 2.4

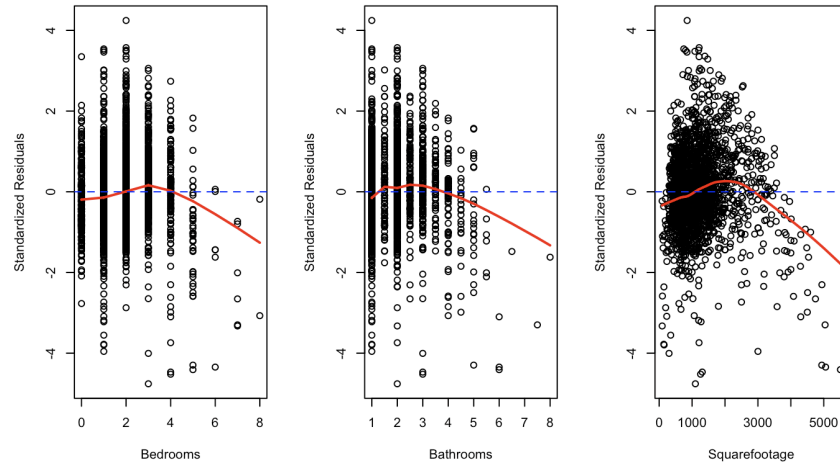


Figure 2.6

|                  | df<br><dbl> | AIC<br><dbl> |
|------------------|-------------|--------------|
| model_log_logsf  | 22          | 425.8573     |
| model_log_sqrtsf | 22          | 335.9945     |

2 rows

|                  | df<br><dbl> | BIC<br><dbl> |
|------------------|-------------|--------------|
| model_log_logsf  | 22          | 551.2837     |
| model_log_sqrtsf | 22          | 461.4209     |

2 rows

Table 2.5

Table 2.6

```
Call:
lm(formula = log(price) ~ zipcode + bedrooms + bathrooms + sqrt(squareFootage) +
    property_type, data = my_data1_clean)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.31975 -0.15588 -0.01488  0.15178  1.13186
```

```
Coefficients:
(Intercept)      6.468912  0.030638 211.137 < 2e-16 ***
zipcode00024    -0.281841  0.026827 10.506 < 2e-16 ***
zipcode00025    -0.159311  0.025339  6.287 3.89e-10 ***
zipcode00034    -0.186338  0.026402  7.058 2.26e-12 ***
zipcode00035    -0.104759  0.027201  3.851 0.000121 ***
zipcode00036    -0.168255  0.030196  5.572 2.83e-08 ***
zipcode00048    -0.256567  0.027641  9.282 < 2e-16 ***
zipcode00049    -0.282193  0.030492  9.255 < 2e-16 ***
zipcode00064    -0.182332  0.032819  5.556 3.10e-08 ***
zipcode00066    -0.214550  0.030507  7.033 2.70e-12 ***
zipcode00069    -0.484827  0.025781 18.806 < 2e-16 ***
zipcode00230    -0.151204  0.042232  3.580 0.000351 ***
zipcode00403    -0.347223  0.033850 10.258 < 2e-16 ***
zipcode00404    -0.261255  0.039283  6.651 3.68e-11 ***
zipcode00405    -0.393255  0.036250 10.848 < 2e-16 ***
bedrooms        -0.037933  0.000210  4.119 3.95e-05 ***
bathrooms        -0.056444  0.010632  5.309 1.21e-07 ***
sqrt(SquareFootage) 0.040450  0.001199 33.741 < 2e-16 ***
property_typeCondo 0.178039  0.016240 10.963 < 2e-16 ***
property_typeSingle 0.168628  0.015924 10.589 < 2e-16 ***
property_typeTownhouse 0.124919  0.020275  6.161 8.57e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.2597 on 2190 degrees of freedom
Multiple R-squared:  0.8067, Adjusted R-squared:  0.8049
F-statistic: 456.9 on 20 and 2190 DF, p-value: < 2.2e-16
```

Figure 2.7